

Team 12: Close The Loop - Tasks

Problem:

- There is a lack of transparency into tasks tracked for patients. Unexpected tasks, if not prioritised, can be lethal.
- There is no system in place that can speak to patients at home on a regular basis following up with their symptoms.
- Existing software keeps a track of tasks requested by the patients while it should be otherwise.

Basic Functionality:

- Input the patient's records (documents, notes, clinical requests/orders and results)
- From the free-text notes and labs:
 - Extract tasks that need to be actioned.
 - These include reading "Book follow up appointment in 4 weeks" or "Chase blood culture results" from the free-text notes. -> Could map free text into a task tag from a bank of potential tasks.
 - Some of these tasks can be inferred from incomplete actions in the EHR, such as a blood test with no result yet. (That means a clinician needs to wait until the results are released and then act on it)
 - Extract the timestamps and values from each entry and recreate a clinical workflow timeline. (Ledger + specific task schema)

Goals:

- Be able to parse the fragmented EHR data sources for clinical tasks, surface unexecuted tasks and display these in a dashboard along with key metrics of the workflow latency.
 - Have an effectuator agent that can action or reduce the friction for task completion -> organise follow up or request a blood test, book a review call itself.
 - Step latency information allows admin to see the critical bottlenecks and time-sinks in each clinical workflow.
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- Latency of workflow steps -> surface
 - ChronJobs
 - CloseTheLoop
 - Compared against future actions and free-text

Ways to improve:

- NICE Guidelines
- BNF Check
- Latent decision making
 - Gather actions for similar patients across different hospital visits
- SNOMED ->

- Predictive
- Coordinate
- QOF
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Schema:

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{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ClinicalTaskEpisode",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "episode_id",
    "timestamp",
    "task_id",
    "status",
    "owner",
    "note",
    "deadline",
    "clinical_priority",
    "snomed_id",
    "performed_by",
    "sample_id",
    "requested_id"
  ],
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    "episode_id": {
      "type": "string"
    },
    "timestamp": {
      "type": "string",
      "format": "date-time"
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    "task_id": {
      "type": "string"
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    "status": {
      "type": "string"
    },
    "owner": {
```

```
    "type": "string",
    "description": "Clinician identifier"
  },
  "note": {
    "type": "string"
  },
  "deadline": {
    "type": "string",
    "format": "date-time"
  },
  "clinical_priority": {
    "type": "string",
    "enum": ["emergency", "urgent", "standard"]
  },
  "snomed_id": {
    "type": "string",
    "description": "SNOMED CT concept identifier"
  },
  "performed_by": {
    "type": "string",
    "description": "Clinician identifier"
  },
  "sample_id": {
    "type": "string"
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  "requested_id": {
    "type": "string",
    "description": "Clinician identifier"
  }
}
```